

Francisco Mondaca

Data engineer responsible for
designing and implementing archive
infrastructure for audio-visual research
data using object storage technologies.
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EDUCATION

- University of Cologne** Cologne, Germany
M.A. in Computational Linguistics and General Linguistics, passed with distinction 2012–2015
– Thesis: “Management and Quality Control of Digital Dictionaries: Enhancing Print Editions through Crowdsourcing”
- University of Cologne** Cologne, Germany
B.A. in Computational Linguistics and General Linguistics 2008–2011
– Thesis: “Linguistische Anwendungen in Mobile Computing: Ein Übersetzungsdienst als Android-App”
- Universidad de Chile** Santiago, Chile
Law 2003–2006

EXPERIENCE

- IT Center University of Cologne** Cologne, Germany
Research Associate June 2024 - Present
– KA3-OS - Cologne Center for Analysis and Archiving of Audio-Visual Data - DFG-funded
– Data engineer responsible for designing and implementing an archive infrastructure for linguistic research data using object storage technologies.
- Cologne Center for eHumanities, University of Cologne** Cologne, Germany
Research Associate June 2021 - May 2024
– Middle Persian Corpus and Dictionary (MPCD) - DFG funded
– Software architect and technical lead overseeing the development of an online collaborative editor, integral to compiling a corpus of Zoroastrian Middle Persian texts in Pahlavi script. This editor, focusing on UD annotations and display, supports the project’s objective of enriching Iranian studies in the context of Sasanian and early Islamic periods, ultimately contributing to creating a Middle Persian-English dictionary.
- Prometheus Bildarchiv, University of Cologne** Cologne, Germany
Research Associate October 2020 - May 2021
– Image Search for the Prometheus Image Archive - NFDI funded
– Responsible for developing and implementing an image search system involving a pipeline for managing nearly 2.8 million images on Prometheus. The approach included generating image embeddings via a neural network and facilitating their clustering and retrieval through Elasticsearch.
- Cologne Center for eHumanities, University of Cologne** Cologne, Germany
Research Associate July 2017 - December 2020
– VedaWeb - DFG funded
– Responsible for data modeling and curation, contributed to software development and architecture, and formulated and implemented a major part of the technical strategy that secured the project’s grant extension.

- Annotierte Rätoromanische Chrestomathie - DFG Funded
- Led backend development and coordinated team efforts
- Anglizismen im Deutschen - DFG Funded
- Oversaw hardware infrastructure setup, backend development, and team coordination
- Pledari Grond - Funded by Lia Rumantscha
- Developed a Rumantsch Grischun conjugator and managed team coordination

Research Assistant

June 2013 - December 2013

- Annotierte Rätoromanische Chrestomathie - DFG funded
- Data transformation
- Pledari Grond - Funded by Lia Rumantscha
- Data curation

Linguee Spanish, Linguee GmbH (DeepL SE)

Cologne, Germany

Translator and Advisor

February 2010 - May 2013

- Head of Linguee Spanish, responsible for initiating and structuring the service. Directly handled data curation and data quality. Managed press relations.
- Responsible for Linguee's expansion to Latin America: Argentina, Chile, Colombia, Mexico and Peru
- Development of Linguee's first Android Application

Pablo Neruda Foundation

Santiago, Chile

January 2005 - March 2007

- English-Spanish tourist guide at La Chascona Museum House
- Copyright analysis of Pablo Neruda's contracts

SELECTED PUBLICATIONS

- [1] **Mondaca, Francisco**, F. Rau, P. Schildkamp, and L. Günther, "The Kosh Suite: A Framework for Searching and Retrieving Lexical Data Using APIs", in *Electronic lexicography in the 21st century (eLex 2023): Invisible Lexicography. Proceedings of the eLex 2023 conference. Brno, 27–29 June 2023*, M. Medveď, M. Měchura, C. Tiberius, I. Kosem, J. Kallas, M. Jakubíček, and S. Krek, Eds., Brno: Lexical Computing CZ s.r.o., 2023, pp. 236–247.
- [2] **Mondaca, Francisco**, K. Rezaia, S. Čéplö, and C. Neufeind, "Towards a Comprehensive Dictionary of Middle Persian", in *Electronic lexicography in the 21st century (eLex 2023): Invisible Lexicography. Proceedings of the eLex 2023 conference. Brno, 27–29 June 2023*, M. Medveď, M. Měchura, C. Tiberius, I. Kosem, J. Kallas, M. Jakubíček, and S. Krek, Eds., Brno: Lexical Computing CZ s.r.o., 2023, pp. 220–35.
- [3] **Mondaca, Francisco** and F. Rau, "Transforming the Cologne Digital Sanskrit Dictionaries into Ontolex-Lemon", in *Proceedings of the 7th Workshop on Linked Data in Linguistics (LDL-2020)*, 2020, pp. 11–14.
- [4] **Mondaca, Francisco**, F. Rau, C. Neufeind, B. Kiss, D. Kölligan, U. Reinöhl, and P. Sahle, "C-SALT APIs - Connecting and Exposing Heterogeneous Language Resources", in *Digital Humanities 2019*, 2019.
- [5] **Mondaca, Francisco**, P. Schildkamp, and F. Rau, "Introducing Kosh, a Framework for Creating and Maintaining APIs for Lexical Data", in *eLex 2019*, 2019, pp. 907–921.

TEACHING

- **Digital Lexicography** - Computational Linguistics, University of Cologne WS 2018/2019
- **Java Training** - Computational Linguistics, University of Cologne SS 2014 - WS 2015/2016

MAIN TECH SKILLS

- **Programming languages:** Python, Java, Javascript
- **Python web frameworks:** Django, Flask
- **Javascript framework:** React
- **Databases:** PostgreSQL, MongoDB
- **Machine Learning / Data Science frameworks:** pandas, scikit-learn, PyTorch, spaCy, Stanza.
- **Search engine:** Elasticsearch
- **Virtualization/Containerization:** Docker, VirtualBox, VMWare
- **WebAPIs:** REST, GraphQL

LANGUAGES

- **English:** C2
- **German:** C2
- **Spanish:** Native Speaker

CAREER BREAKS

Parental leave

K1

K2

February 2016 - August 2016

February 2018 - September 2018